

DATA REPRESENTATION PRACTICE SHEET

NIMCET TARGET PAPER • CORE FUNDAMENTALS (LEVEL: EASY)

ResourceOS

1. How many total distinct character codes can standard 7-bit ASCII represent?

- (A) 64
- (B) 128
- (C) 256
- (D) 512

2. How many binary bits correspond to a single hexadecimal digit?

- (A) 2 bits
- (B) 3 bits
- (C) 4 bits
- (D) 8 bits

3. What is the range of signed integers that can be stored in an 8-bit 2's complement representation?

- (A) -128 to +127
- (B) -127 to +127
- (C) -128 to +128
- (D) 0 to 255

4. Which representation scheme has two distinct binary encodings for zero (+0 and -0)?

- (A) 2's Complement
- (B) Sign-Magnitude
- (C) Unsigned Binary
- (D) Excess-128 notation

5. How is the 1's complement of a binary number formed?

- (A) Adding 1 to the least significant bit
- (B) Inverting all bits (changing 0s to 1s and 1s to 0s)
- (C) Shifting all bits left by one position
- (D) Inverting only the Most Significant Bit (MSB)

6. What is the 2's complement of the binary number 1010?

- (A) 0101
- (B) 0110
- (C) 1011
- (D) 0100

7. What does the hexadecimal symbol 'F' represent in decimal?

- (A) 10
- (B) 12
- (C) 15
- (D) 16

8. Which standard character encoding uses up to 32 bits and can represent characters from almost all spoken languages?

- (A) ASCII
- (B) EBCDIC
- (C) BCD
- (D) Unicode

9. In an IEEE 754 Single-Precision Floating-Point standard, how many total bits are used?

- (A) 16 bits
- (B) 32 bits
- (C) 64 bits
- (D) 80 bits

10. In IEEE 754 32-bit floating-point format, how many bits are allocated for the Exponent field?

- (A) 1 bit
- (B) 8 bits
- (C) 23 bits
- (D) 11 bits

11. What is the binary equivalent of the decimal fraction 0.5?

- (A) 0.12
- (B) 0.1
- (C) 0.01
- (D) 0.05

12. What decimal value does the fractional binary number 0.01 represent?

- (A) 0.5
- (B) 0.25
- (C) 0.125
- (D) 0.1

13. In signed numbers, what does the Most Significant Bit (MSB) equal to 1 indicate?

- (A) Positive number
- (B) Negative number
- (C) Zero
- (D) Fractional overflow

14. What is the bias value used in the exponent field of IEEE 754 Single Precision format?

- (A) 63
- (B) 127
- (C) 255
- (D) 1023

15. What is the general formula for the maximum unsigned decimal integer that can be represented with n bits?

- (A) 2^n
- (B) $2^n - 1$
- (C) 2^{n-1}
- (D) $2^{n+1} - 1$

16. EBCDIC is an 8-bit character encoding format developed primarily by which organization?

- (A) Microsoft
- (B) IEEE
- (C) IBM
- (D) ANSI

17. Which binary code is unweighted and changes by only one bit between consecutive values?

- (A) BCD Code
- (B) Gray Code
- (C) Excess-3 Code
- (D) ASCII Code

18. What is the binary representation of decimal number 13?

- (A) 1100
- (B) 1101
- (C) 1011
- (D) 1110

- 19. What is the primary advantage of 2's complement over 1's complement for arithmetic logic units?**
- (A) It requires fewer bits to store numbers. (B) It eliminates end-around carry and provides a unique representation for zero.
- (C) It supports characters along with numbers. (D) It automatically detects parity errors.
- 20. What is the decimal value of the binary fraction 0.11?**
- (A) 0.33 (B) 0.75
- (C) 0.5 (D) 0.625
- 21. How many octal digits are needed to represent a 12-bit binary number?**
- (A) 3 (B) 4
- (C) 6 (D) 2
- 22. In floating-point numbers, what does "normalization" ensure?**
- (A) The leading bit of the non-zero mantissa is non-zero (typically 1). (B) The exponent is always positive.
- (C) The sign bit is always set to zero. (D) The fraction has an even number of 1s.
- 23. What happens when an addition operation on two positive 8-bit signed numbers yields a result greater than +127?**
- (A) Underflow (B) Arithmetic Overflow
- (C) Syntax Error (D) Zero division fault
- 24. In BCD (Binary Coded Decimal), how many bits are used to represent each decimal digit (0 to 9)?**
- (A) 2 bits (B) 3 bits
- (C) 4 bits (D) 8 bits
- 25. How many bits are allocated for the mantissa (fraction) field in IEEE 754 Double Precision (64-bit)?**
- (A) 23 bits (B) 32 bits
- (C) 52 bits (D) 63 bits

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(B)	11	(B)	16	(C)	21	(B)
2	(C)	7	(C)	12	(B)	17	(B)	22	(A)
3	(A)	8	(D)	13	(B)	18	(B)	23	(B)
4	(B)	9	(B)	14	(B)	19	(B)	24	(C)
5	(B)	10	(B)	15	(B)	20	(B)	25	(C)